MATH 243 — Probability and Statistics

Analysis of Variance: F-test for Two Variances, One-way and Two-way ANOVA

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Quick Notes — ANOVA — revision bullets

NOTE Page numbers refer to 8. Lecture- 12- ANOVA(KKS Sir_s Part).pdf (10 pp.), written as ANOVA p. n. Critical values come from 8.1 F-Distribution-Table.pdf and 8.2 t-table.pdf. Two-way ANOVA (asked 2022-23 Q8(c)) is not on the slides; Lesson 19 teaches it from the textbook (marked). "TF p." = merged question PDF page; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.4	Q8(b) Retail prices in 3 cities (4 shops each): one-way ANOVA (F0.05(2, 9) = 4.26) (20) [L18]
2022-23 (MATH 243) TF p. 51	Q8(b) Lifetimes of 4 brands of bulbs (3 each): one-way ANOVA at 5 % and 1 % (10) [L18] · Q8(c) Four corn types in five blocks: two-way ANOVA for blocks and corn types (5 %) (15) [L19]
2021-22 (MATH 245) TF p. 45	Q8(b) Salesmen A, B, C weekly sales: one-way ANOVA (F(2,11) = 3.98) (18) [L18]

EXAM FOCUS 4 questions, 63 marks, all since 2021-22: one-way ANOVA (18–20 marks) in three consecutive papers, plus a two-way ANOVA in 2022-23. The computation is mechanical — G, CT, ST, SB, SW, the ANOVA table, compare F with the table value. Practise it until it takes ten minutes, and memorise the two-way extension.

Lesson 18 — F-test for Two Variances and One-way ANOVA

ANOVA pp. 1–10

1. Comparing Two Population Variances (ANOVA pp. 1–3)

Assumptions: independent samples; normal populations; random sampling; quantitative (interval/ratio) data; under H_0 the variances are equal.

$$H_0: \sigma_1^2 = \sigma_2^2; \quad F = S_1^2/S_2^2 \text{ (larger sample variance on top)}$$
$$\text{Reject } H_0 \text{ if } F \geq F_{\alpha,\,(n1-1),\,(n2-1)} \text{ (one-tailed) or } F_{\alpha/2,\,\ldots} \text{ (two-tailed)}$$

Lammers Limos: U.S. 25 (n = 7) $\bar{x}$ = 58.29, s = 8.9947; I-75 (n = 8) $\bar{x}$ = 59.00, s = 4.3753. F = $8.9947^2/4.3753^2 = 4.23 > F_{0.05,\,6,\,7} = 3.87$ (two-tailed at α = 0.10) → reject H_0: the variation in driving time differs.

Exercises (ANOVA pp. 3–4) — answers (s^2 with divisor n − 1)

Exercise	s^2 (first / second)	F	Critical	Decision
two-tailed α = 0.05	6 / 3.3333	1.8	F(7,9) = 4.2	do not reject
H1: $\sigma_A^2 > \sigma_B^2$, α = 0.05	56.2381 / 1.5	37.4921	F(6,8) = 3.58	reject H0
H1: $\sigma_B^2 > \sigma_A^2$ (A more consistent), α = 0.05	11.7143 / 0.5667	20.6723	F(7,5) = 4.88	reject H0

2. One-way ANOVA (ANOVA pp. 4–6)

k independent samples from k normal populations with a common variance σ^2. H_0: $\mu_1 = \ldots = \mu_k$; H_1: at least two means differ.

$$G = \Sigma\Sigma\, x_{ij}, \quad N = \Sigma\, n_i, \quad CT = G^2/N$$
$$ST = \Sigma\Sigma\, x_{ij}^2 - CT, \quad SB = \Sigma\, T_i^2/n_i - CT, \quad SW = ST - SB$$
$$F = [SB/(k-1)] \,/\, [SW/(N-k)] \sim F_{(k-1),\,(N-k)}$$

Source	SS	d.f.	MS	F
Between	SB	k − 1	MSB = SB/(k−1)	MSB/MSW
Within	SW	N − k	MSW = SW/(N−k)	
Total	ST	N − 1		

Reactor example (ANOVA pp. 6–9)

T = 375, 344, 423; means 75, 68.8, 84.6; G = 1142; N = 15; CT = $1142^2/15$ = 86944.27
SB = $375^2/5 + 344^2/5 + 423^2/5$ − CT = 633.73
Σx^2 = 87,716; ST = 771.73; SW = 138
F = (633.73/2)/(138/12) = 316.87/11.5 = 27.55 > $F_{0.05,2,12}$ = 3.89 → reject H0

NOTE The slide gives Σx^2 = 87713, ST = 768.73, SW = 135.20 and F = 28.10. Adding the 15 squares gives 87716, so ST = 771.73, SW = 138.00, MSE = 11.5 and F = 27.55. The decision does not change. Its pairwise t values (2.91, 4.51, 7.42) use MSE = 11.27; with 11.5 they are 2.89, 4.48, 7.37 — all still > 2.179.

3. Pairwise Comparisons (ANOVA pp. 8–9)

$$t = |\bar{x}_i - \bar{x}_j| \,/\, \sqrt{(MSE\,(1/n_i + 1/n_j))}, \quad MSE = SW/(N-k)$$

Reject $\mu_i = \mu_j$ if $|t| \geq t_{\alpha/2,\, N-k}$

Exam Approach

1. State H0, H1 and α. Compute T_i, n_i, G, N, CT.
2. ST from Σx^2 (code the data, e.g. divide by 100, if numbers are large — F does not change).
3. SB from $\Sigma\, T_i^2/n_i$; SW = ST − SB; build the ANOVA table with d.f. and mean squares.
4. F = MSB/MSW; compare with $F_{\alpha,(k-1),(N-k)}$; conclude in words. Pairwise t only if H0 is rejected.

Practice — Lesson 18

(a) Actual — 2021-22 Q8(b) (TF p. 45). (b) Actual — 2022-23 Q8(b) (TF p. 51). (c) Actual — 2023-24 Q8(b) (22 batch p. 4).

(a) Salesmen A, B, C: weekly sales (Tk thousand) A: 300, 400, 300, 500, 0; B: 600, 300, 300, 400; C: 700, 300, 400, 600, 500. Test whether the average sales differ ($F_{(2,\,11)}$ at 5 % = 3.98). (18)

(b) Lifetimes of four brands of bulbs: A 20, 19, 21; B 25, 23, 21; C 24, 20, 22; D 23, 20, 20. Test at 5 % and 1 % whether the mean lifetimes differ. (10)

(c) Retail prices (Tk) of a commodity in shops of three big cities: A 15, 18, 23, 24; B 23, 19, 18, 24; C 19, 16, 22, 31. Carry out the analysis of variance to decide whether prices differ between the cities. Do the average prices differ? ($F_{0.05}(2, 9) = 4.26$) (20)

✓ Solution

(a) Code x/100: A 3, 4, 3, 5, 0 (T = 15); B 6, 3, 3, 4 (T = 16); C 7, 3, 4, 6, 5 (T = 25); N = 14, G = 56.

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CT = 56²/14 = 224;  Σx² = 59 + 70 + 135 = 264;  ST = 264 − 224 = 40
SB = 15²/5 + 16²/4 + 25²/5 − 224 = 45 + 64 + 125 − 224 = 10;  SW = 40 − 10 = 30
(in original units multiply sums of squares by 100² = 10,000: SB = 100,000, SW = 300,000)
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Source	SS (Tk thousand²)	d.f.	MS	F
Between	100,000	2	50,000	1.833
Within	300,000	11	27272.73	
Total	400,000	13		

F = 1.833 < $F_{0.05}(2, 11) = 3.98$ → do not reject H_0: no significant difference in average sales.

(b) T = 60, 69, 66, 63; G = 258, N = 12, CT = 258²/12 = 5547; $\Sigma x^2 = 5586$.

Source	SS	d.f.	MS	F
Between	15	3	5	1.667
Within	24	8	3	
Total	39	11		

F = 1.667 < $F_{0.05}(3, 8) = 4.07$ and < $F_{0.01}(3, 8) = 7.59$ → the mean lifetimes do not differ significantly at either level.

(c) T = 80, 84, 88; G = 252, N = 12, CT = 5292; $\Sigma x^2 = 5506$.

Source	SS	d.f.	MS	F
Between	8	2	4	0.175
Within	206	9	22.89	
Total	214	11		

F = 0.175 < 4.26 → do not reject H_0; the average prices in the three cities are the same.

Lesson 19 — Two-way ANOVA (One Observation per Cell)

textbook — asked 2022-23 Q8(c)

r blocks (rows) × c treatments (columns), one observation per cell. Two hypotheses: no block effect and no treatment effect.

$$CT = G^2/N,\ \ ST = \Sigma x^2 - CT,\ \ SR = \Sigma R_i^2/c - CT,\ \ SC = \Sigma C_j^2/r - CT,\ \ SE = ST - SR - SC$$

$$F_{rows} = [SR/(r-1)]/[SE/((r-1)(c-1))],\ \ F_{cols} = [SC/(c-1)]/[SE/((r-1)(c-1))]$$

Exam Approach

1. Row totals R_i, column totals C_j, grand total G, N = rc.
2. CT, ST, SR, SC, SE as above; d.f. r − 1, c − 1, (r − 1)(c − 1), N − 1.
3. Two F ratios, each against F_α with its own numerator d.f. and the error d.f.; conclude for blocks and for treatments.

Practice — Lesson 19

Actual — 2022-23 Q8(c) (TF p. 51).

Yields of four types of corn I–IV in five blocks A–E: A 12, 15, 10, 14; B 15, 19, 12, 11; C 14, 18, 15, 12; D 11, 16, 12, 16; E 16, 17, 11, 14. Test at 5 % whether there are differences due to blocks and due to corn types. (15)

✓ Solution

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Block totals R: 51, 57, 59, 55, 58;  corn totals C: 68, 85, 60, 67;  G = 280, N = 20, CT = 280²/20 = 3920
Σx² = 4044 → ST = 124
SR = (51² + 57² + 59² + 55² + 58²)/4 − 3920 = 10
SC = (68² + 85² + 60² + 67²)/5 − 3920 = 67.6
SE = 124 − 10 − 67.6 = 46.4
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Source	SS	d.f.	MS	F	$F_{0.05}$
Blocks	10	4	2.5	0.647	3.26
Corn types	67.6	3	22.533	5.828	3.49
Error	46.4	12	3.867		
Total	124	19			

Blocks: F = 0.65 < 3.26 → no significant block difference. Corn types: F = 5.83 > 3.49 → the corn types differ significantly at 5 %.

Quick Notes — ANOVA

Revision points for Lessons 18–19.

L18 · F-test and One-way

- F = larger s^2/smaller s^2; two-tailed uses α/2. Limos 4.23 > 3.87.
- CT = G^2/N; ST = $\Sigma x^2 - CT$; SB = $\Sigma T^2/n - CT$; SW = ST − SB; F = MSB/MSW.
- Reactor F = 27.55 (slide 28.10 from a Σx^2 slip). Salesmen 1.83 < 3.98; bulbs 1.67 < 4.07; cities 0.17 < 4.26.
- Pairwise $t = |\bar{x}_i - \bar{x}_j|/\sqrt{(MSE(1/n_i + 1/n_j))}$ vs $t_{\alpha/2,\, N-k}$.

L19 · Two-way

- SR = $\Sigma R^2/c - CT$, SC = $\Sigma C^2/r - CT$, SE = ST − SR − SC; error d.f. $(r - 1)(c - 1)$.
- Corn: blocks F = 0.65 (n.s.), types F = 5.83 > 3.49 (significant).